

ROBERTO GUAGNELLI GARCÍA

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Mechanical Engineering Student | Overall Aircraft Design & Systems Integration

Spanish & Mexican citizen | EU work authorization | Open to relocation

EDUCATION

Tecnológico de Monterrey, Campus Estado de México

Expected Jul 2027

B.S. in Mechanical Engineering; Concentration in Aeronautics

Academic average: 91/100

WORK EXPERIENCE

GE Aerospace | Designer Intern, Product Definition

Aug 2026 – Present

RISE Open Fan Program

- Design brackets and harness routing in Siemens NX: 70+ brackets and 20+ routing revisions, applying manufacturing and maintainability criteria.
- Resolve clearance, interference and minimum bend radius issues through harness routing and support-bracket revisions.
- Prepare product definition drawings and tolerance tables using established datums for brackets and harnesses.

GE Aerospace | Lean Summer Experience Intern

Jun 2026 – Jul 2026

Integrated Performance & Operability (IP&O)

- Implemented visual planning and KPI dashboards with program leads; reduced meeting duration by 20%, as timed by meeting leaders.

ENGINEERING PROJECTS & LEADERSHIP

Borregos CEM AeroDesign | Tecnológico de Monterrey

Sep 2023 – Present

Engineering & Leadership Advisor

Sep 2026 – Present

- Mentor the incoming Design Captain and advise engineering groups and team leadership.

Engineering Coordinator / Design Captain

Aug 2024 – Sep 2026

- Led 5 designers and coordinated aircraft integration across a 40+ member team; documented requirements and conducted multidisciplinary reviews.
- Resolved a structural/electrical interface conflict by coordinating geometry and cable-passage changes.
- Modeled 300+ CAD parts per aircraft in 400+ component assemblies; evaluated mass, CG, stability and propulsion trade-offs.
- Designed removable joints for Wingull's 4.2 m wing, keeping components under 1.2 m; supported its 18 kg distributed-load test.
- Used steady/transient CFD and XFLR5/AVL to assess aerodynamics, stability, trim and control authority; applied drag results to takeoff and payload estimates.

Manufacturing Coordinator | Hyperion

Sep 2023 – Aug 2024

- Coordinated 8 members in fabrication, assembly and quality control; designed and built a spring-loaded aluminum nose gear for limited propeller clearance.
- Helped build Hyperion: 6 competition flights with 3 kg payload at 4 kg empty mass; 2nd overall and 3rd in design report at SAE Aero Design México 2024.

TECHNICAL SKILLS

CAD & Analysis: Siemens NX, SolidWorks, Fusion 360, ANSYS Fluent, XFLR5, AVL; steady-state and transient CFD.

Programming & Data: Python, MATLAB, Excel, Power BI.

Aircraft & Systems: Aircraft design, systems integration, CG, propulsion selection, requirements, interfaces, verification/testing.

Stability & Control: Static stability, trim, control derivatives and control authority using AVL/XFLR5.

Manufacturing: 3D printing, carbon-fiber fabrication, laser cutting, aluminum welding, tube and sheet bending.

Certification: Certified SOLIDWORKS Professional (CSWP), perfect exam score.

Languages: Spanish (native), English (C2; Cambridge and IELTS), German (currently learning at A2 level).